Kalan Brunell

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Education

Tufts University, Medford, MA — Expected May 2027
Bachelor of Science in Computer Engineering — *Dean's List*

Experience

Manufacturing Automation Engineering Intern, Gentex Corporation - Manchester, NH — Summer 2025

- Designed a custom 2D inspection system to increase manufacturing inspection accuracy by up to 60%.
- Enabled real-time object location and automated quality inspections by integrating the vision system with manufacturing robotic arms via a custom microcontroller-based protocol converter.
- Developed Fanuc robot programs using the vision system to automate manufacturing tasks like soldering, adhesive application, and part manipulation.
- Performed validation and bench debugging using oscilloscopes, logic analyzers, and controlled test fixtures to validate sensor timing, signal integrity, and system reliability.

Projects

FSAE Electric Racecar Accumulator, Tufts Electric Racing - Project Manager — May 2025 - Present

- Leading the design of a removable high-voltage battery pack for a Formula SAE Electric racecar.
- Designing real-time battery management and monitoring firmware on an embedded STM32 microcontroller.
- Developed PCBs including real-time cell and thermistor monitoring, power distribution and safety systems.
- Implemented fault detection, isolation, and safety interlocks aligned with real-time embedded reliability requirements.

MITRE Embedded Capture the Flag (eCTF), Tufts University — January 2026 - Present

- Competing in an embedded systems security competition to design, analyze, and exploit an embedded device.
- Implemented secure firmware features on an ARM Cortex microcontroller to mitigate hardware-based vulnerabilities such as UART buffer overflows and side-channel attack vectors.
- Analyzed attack surfaces including UART interfaces, flash memory access, and fault injection vectors, to identify and take advantage of security weaknesses in opposing teams' devices.

Autonomous Color Following Vehicle, Tufts University — September 2025 - December 2025

- Developed an autonomous/WebSocket controllable vehicle to navigate via color and object detection powered by an embedded ATmega MCU.
- Created a purpose-built photodiode array and C++ algorithm with embedded ADCs to detect and follow course markers via PID control.
- Implemented WebSocket communication to allow real-time telemetry, remote control, and interaction with other on-field vehicles.

FIRST Robotics, Lead Engineer, Captain — August 2022 - May 2023

- Developed a retroreflective-based vision system which calculated real-time distance and angle to target.
- Programmed field-relative swerve drive and autonomous control architecture in WPILib, integrating IMU and encoder feedback for precise odometry and path tracking.
- Integrated sensors, actuators, and motor controllers into robust electromechanical subsystems, enabling closed-loop control.

Skills

Programming: C, C++, Python, ARM Assembly, RISC, CMSIS, RTOS, OOP, Scripting
Embedded & Electronics: VHDL, Waveform Debugging, GDB, Digital Logic, FPGAs, I2C/UART/CAN, GPIO, ADC
Circuit & PCB Design: Altium Designer, KiCad, LTspice, Electrical Lab Equipment, Soldering
Robotics & Automation: OpenCV, Limelight, Robotic Control, PLC Programming
Tools: Lattice Radiant, SolidWorks, Onshape, Rapid Prototyping, 3D Printing, MATLAB, GIT, LaTeX